

The last 75 years has seen many firsts in the financial world: credit card in 1950 (Diners Club); automated teller machine (ATM) in 1967 (Barclays Bank); telephone banking in 1983 (Bank of Scotland); Internet banking in 1994 (Stanford Federal Credit Union); radio-frequency identification (RFID) payments in 1997 (Mobil Speedpass); chip-and-pin credit cards in 2005 (Mastercard); and Apple Pay with a mobile device in 2014 (Apple).

Importantly, all these innovations were built on the backbone of centralized finance. While there have been some technological advances, the structure of today's banking system has not changed much in the past 150 years. That is, digitization still supported a legacy structure. The high costs associated with this legacy system has spurred further advances known as *fintech*.

FINTECH

When costs are high, innovation will arise to capitalize on inefficiencies. Sometimes, however, a powerful layer of middle people can slow this process. An early example of decentralized finance emerged in the foreign currency (forex) market 20 years ago. At the time, large corporations used their investment banks to manage their forex needs. For example, a U.S.-based corporation might need €50 million at the end of September to make a payment on some goods purchased in Germany. Its bank would quote a rate for the transaction. At the same time, another client of the bank might need to sell €50 million at the end of September.